

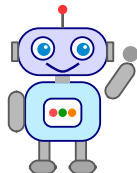
Writing a Good Journal or Conference Paper

... and using ChatGPT & other AI tools without getting burned

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October 2026



- ▶ Your project report is **not** a paper. A paper is an **argument** for one idea, backed by evidence, aimed at a busy stranger.
- ▶ A paper is a career asset: higher studies, jobs, research internships, patents.
- ▶ Most first submissions fail for **avoidable** reasons: no clear argument, weak structure, wrong venue, careless writing.
- ▶ AI tools are now part of the workflow. Used well, they save time. Used carelessly, they can **end your paper** (fake references, policy violations, plagiarism).

By the end you should be able to...

1. State your paper's argument in two sentences
2. Pick a credible venue (and avoid predatory ones)
3. Draft and revise each section with a method
4. Handle submission, reviews and rejection
5. Use AI tools **safely and honestly**

Framework adapted from: Belcher, *Writing Your Journal Article in Twelve Weeks*, 2nd ed. (2019); Kallestinova, "How to Write Your First Research Paper," *Yale J Biol Med* 84(3):181–190 (2011).

- ▶ Reviewers **skim first** and decide whether to read closely. Writing advisor Matt Might: abstract, intro and conclusion are “80% of your paper from the perspective of a referee.”
- ▶ Readers look at about **1–3 pages** of an article online. Many never read past the first page.
- ▶ Your **title** is read $\sim 1000\times$ more than your article.
- ▶ Many people cite from the abstract alone. One study estimated only $\sim 20\%$ of citers had read the paper they cited.

So, spend your effort where it counts

Title → abstract → introduction → conclusion must tell **one clear story**, even to a distracted reader.

Write for the skimmer

State your claim early. Make assumptions explicit. Never make the reader hunt for the contribution.

- 1 What actually gets a paper published
- 2 Choosing a venue: journal vs. conference, predatory traps
- 3 Writing the paper: outline, sections, abstract, title
- 4 Revising: macro, micro, and editing diagnostics
- 5 Submitting, surviving peer review, and a 12-week plan
- 6 **Using ChatGPT & other AI tools: what to do, what to avoid**
- 7 Final checklist and take-aways

PART 1

What gets published

What makes a paper worth publishing?



Myth

“Only profoundly theoretical or ground-breaking work gets published.”

“Lots of interesting ideas make a paper publishable.”

“It must be entirely original.”

Reality (Belcher)

Most published articles are **narrow** in claim and context, written by people like you and me.

Papers are published for articulating **one important idea**, not for spraying many. Ideas must be organised around one claim.

Almost all scholarly publications are built on earlier work. You need something **new**, and visibly **related to the old**.

The working definition

A publishable article is organised around **one important new idea** that is demonstrably related to previously published work: “something new about something old.”

Narrow and clear beats grand and vague!



Three ways to be “new” (an engineering reading)

Type	What it means	Example in our field
New evidence, old approach	New data/experiments tested against an accepted theory or method	Existing segmentation network evaluated on a new, clinically annotated dataset, <i>and</i> analysed for why it fails (e.g. domain shift)
Old evidence, new approach	New method, design or theory applied to known data	New loss function or architecture benchmarked on standard public datasets
Old + old, new link	Pair two existing things nobody has linked	Classical signal-processing idea (e.g. wavelets) joined with a transformer for a new task

Warning

New evidence alone is a *report*, not a paper. You must **relate it to an ongoing scholarly concern**: “we trained model X on dataset Y and got 91%” is not an argument.

An argument is...

1. your article's **single significant idea**,
2. stated in **one or two sentences**,
3. **early and clearly**,
4. the thing the whole paper is **organised around**,
5. linked to the **existing literature**,
6. and supported by **evidence**.

In science: an argument is simply an *answered research question* or *confirmed hypothesis*.

Belcher (2019), Week 2, pp. 66–68.

Is it an argument? Two quick tests

Agree/disagree test. Can a reasonable person say “I disagree”? If nobody could dispute it, it is a fact, not an argument.

Gut test. Say it aloud. If you feel a little nervous, it probably is an argument.

Toulmin's three parts

Grounds (data) → **Claim** (what you show),
linked by a **warrant** (why the data supports the claim).

Only a topic (weak)

Deep learning for skin lesion segmentation.

A comparison of CNNs and transformers for chest X-rays.

Using ChatGPT in education.

An argument (strong)

Adding a boundary-aware loss to a U-Net improves lesion-edge accuracy on low-contrast images, because.....

Under limited labelled data ($< N$ images), CNNs match transformers on chest X-ray classification at a fraction of the compute; the gap closes only beyond.....

Students who draft first and use LLMs only for language editing retain more conceptual accuracy than students who generate first and edit later.

Template for a technical argument

Method M improves metric Y over baseline B under condition C , because of mechanism Z ; we show this with experiments E .

Main reason: no (or unclear) argument.

- ▶ Editors may say “not original,” “disorganised,” “poor analysis,” or “reads like a student report.” The root cause is often a missing, buried or fuzzy argument.
- ▶ **Even good papers get rejected**: editors reject up to a quarter of submissions without review because the journal has just published something similar, has a backlog, or lacks space.
- ▶ On average, peer review takes **3–4 months**. A wrong venue choice can cost you a year.

Typical technical reasons (instructor's list)

- ▶ Weak or unfair baselines; missing recent comparisons
- ▶ No ablation: which component actually helps?
- ▶ Single run, no variance or significance
- ▶ Data leakage; patient-level splits ignored
- ▶ Not reproducible (no code, seeds, splits, hyper-parameters)
- ▶ Claims larger than the evidence
- ▶ Out of the venue's scope

Can you answer these in one line each?

1. **Problem:** what is broken or unknown?
2. **Gap:** what do existing papers not do?
3. **Idea:** what exactly is new?
4. **Evidence:** which experiments prove it?
5. **So what?** who benefits, and how much?

Evidence you will be asked for

- ▶ Standard datasets + a realistic one, with clear splits
- ▶ Strong, recent, fairly-tuned baselines
- ▶ Ablation study and failure cases
- ▶ Mean $\pm$ SD over several seeds; a significance test
- ▶ Compute cost and limitations
- ▶ Code/data availability

If you cannot fill this in, you do not have a paper yet; you have a project. Go back to the experiments, not to the word processor.

PART 2

Choosing a venue

Pick your venue wisely!



	Conference	Journal
Timeline	Fixed deadline; decision typically in 1–3 months	Rolling submission; multiple rounds, often 3–6+ months
Length / depth	Short (often 4–10 pages); one focused idea	Longer (often 8–20+ pages); fuller theory, experiments, discussion
Benefit	Feedback, networking, presenting	Archival record; often valued for degrees and jobs
Risk	Weak or fake conferences; proceedings may not be indexed	Slow; heavier revision work
Typical use	First results; fast-moving areas (e.g. AI/vision)	Mature, complete studies; extended conference work

Practical advice for a B.Tech project

A good *conference* paper is a realistic first target. Then, if the work is strong, extend it for a journal with substantial new material, and declare the overlap.

- ▶ Sending to “the best journal” without research wastes months. You **cannot submit to two venues at once**: “shotgunning” is an ethical violation and can get you banned.
- ▶ Read **recent issues**: what do they publish, how long, with what depth?
- ▶ Cite and engage with papers from that venue; editors notice.
- ▶ Check: peer review, indexing, publisher, time to decision, fees, open-access terms, article types.

Venue short-list worksheet

Pick 3–5 venues and tick:

- ☐ Scope matches my contribution
- ☐ Peer reviewed (not just editor-read)
- ☐ Indexed (e.g. Scopus / Web of Science / DOAJ / IEEE Xplore / ACM DL)
- ☐ Recent papers of similar type
- ☐ Review time acceptable
- ☐ Fees and licence clear
- ☐ Supervisor approves

Pick the venue before
you write.



Avoid for a research article

- ▶ **Predatory journals:** take fees, little or no real review
- ▶ **News and trade periodicals:** not peer reviewed
- ▶ **Newspapers and magazines:** no citations, no academic weight
- ▶ **Print-only journals:** hardly anyone finds them

Think twice (“debatable”)

- ▶ **Unreviewed proceedings:** check peer review is real
- ▶ **Preprint servers:** fine, but check the target journal's policy (e.g. SHERPA RoMEO)
- ▶ **Edited-volume chapters:** far fewer readers
- ▶ **Non-peer-reviewed journals:** no expert review
- ▶ **Brand-new journals** (under ~3 years): high failure rate

Gold standard: peer reviewed, from a credible publisher, written for scholars, authors cite sources and describe their methods so others can check them.

Red flags

- ▶ Unsolicited email invitation, often with spelling and grammar errors
- ▶ Promises of very fast publication, “no hoops”
- ▶ Free-mail address; no country or publisher details
- ▶ Fee revealed only *after* acceptance
- ▶ Editorial board lists scholars who never agreed (checking the board alone is not enough)
- ▶ Impersonates a real journal and asks for a sudden fee
- ▶ Conference in a tourist city, “all topics,” dozens of tracks

Verify before you submit

- ▶ Type the **official** website yourself; do not click the email link
- ▶ Check indexing on the **indexer's own site** (Scopus source list, Web of Science master list, DOAJ)
- ▶ Check the publisher (IEEE, ACM, Springer, Elsevier, Wiley, . . .) lists the journal or conference itself
- ▶ Ask: do I know anyone who published here and had real reviews?
- ▶ Show the venue to your supervisor first

Integrity rules

- ▶ **One paper, one venue at a time.** No simultaneous submissions.
- ▶ **No duplicate publication.** A journal extension of a conference paper needs substantial new content and a clear disclosure.
- ▶ **No plagiarism**, including self-plagiarism; paraphrasing without citation counts.
- ▶ **Never fabricate or “adjust” data or images.** Selecting is fine; distorting is falsification.

Authorship

- ▶ Decide who is an author **before** writing, and the order. Revisit if roles change.
- ▶ An author contributes substantially to the work, helps draft or revise it, approves the final version, and is **accountable** for it.
- ▶ Supervisors, students and collaborators: agree in writing (email is enough).
- ▶ Ethics approval and consent for human or patient data; dataset licences; conflicts of interest and funding statements.

- ▶ Read the **call for papers** fully: scope, page limit, template (IEEE / ACM / Springer LNCS), anonymity rules.
- ▶ **Double-blind?** Remove names, affiliations, and self-identifying links; cite yourself in the third person.
- ▶ Plan back from the **deadline**: abstract deadlines often come first.
- ▶ Respect the page limit; do not shrink fonts or margins.
- ▶ After reviews: a **rebuttal** window may exist. Reply briefly and politely.
- ▶ **Camera-ready**: implement promised changes; check copyright/licence forms.
- ▶ Presentation: roughly one slide per minute; one message per slide; rehearse.
- ▶ Budget travel and registration *before* you submit; at least one author must usually register and present.

**Deadlines come fast.
Plan backwards!**



PART 3

Writing the paper

Plan it. Draft it. Don't polish yet!



1. Create **regular time blocks** for writing and keep them like appointments.
2. Create a **detailed outline** and discuss it with your mentor and peers.
3. Be meticulous and accurate in **Materials and Methods**; keep one point of view per paragraph.
4. Be **clear, concise and objective** in Results.
5. **Interest the reader** in the Introduction: signal what is new.
6. Present principles and relationships in a **concise, convincing tone** in the Discussion.
7. **Revise** at macro and micro levels with different techniques; get feedback; revise again.

Kallestinova (2011), Rules 1–7. The rest of this section follows her order.

Writing is a skill like exercise

“Writing sessions” are like workouts: regular, short, scheduled. Choose 1–2 hour blocks when you think best; start with the **hardest task first**.

Write on a schedule, like a class.



Rule 1: schedule writing

- ▶ Block 1–2 hours in your calendar and treat them as **non-cancellable appointments**.
- ▶ Mornings work for many people; one student wrote 8–9 a.m. daily for a semester and finished a full draft.
- ▶ Belcher: writing a bit **most days** beats waiting for a free weekend.

Rule 2: start with an outline

- ▶ Begin with visuals (figures, tables, equations, algorithms) and list your **findings**.
- ▶ Expand into Introduction, Methods, Results, Discussion.
- ▶ Talk it through with mentor and peers. It is the cheapest time to fix a weak idea.
- ▶ The outline also helps you pick the **target venue**. Keep a list of 2–3 backups.

Level 1: five questions

1. What is the topic of my paper?
2. Why is this topic important?
3. How could I formulate my hypothesis?
4. What are my results (include visuals)?
5. What is my major finding?

Level 2: group ideas into sections

Introduction: Why is the work important? What is known? Hypotheses? Objectives?

Methods: What materials/data? Who/what were the subjects? What design? What procedure?

Results: Most significant results? Supporting results?

Discussion/Conclusions: Major findings? Significance and implications?

*For ML papers add: **Related Work** (after the Introduction), **Datasets & Metrics**, **Ablations**, **Limitations**.*

- ▶ Copy your outline into a new file and **expand each point**. Pour ideas in; do not polish.
- ▶ Resist fixing words and sentences during drafting. As Paul Silvia puts it, revising while you generate text is like *decaffeinated coffee in the early morning: noble idea, wrong time*.
- ▶ “Writer’s block” is an excuse: your screen is not empty, you have a **template to fill in**.
- ▶ First drafts are **allowed to be bad**.

Suggested writing order

1. Methods (you already have notes)
2. Results (with figures and tables)
3. Introduction
4. Discussion / conclusion
5. Abstract and title (draft early as a thinking tool; finalise last)

Bad first drafts are allowed!



Missing information

“The model was trained on the dataset.”

“Images were pre-processed.”

Reproducible

“A ResNet-50 initialised with ImageNet weights was fine-tuned for 50 epochs (AdamW, lr 10^{-4} , batch 32, cosine decay) on the ISIC-2019 training set, split 70/10/20 at patient level, random seed 42. Images were resized to 224×224 and normalised per channel. Code: [link].”

- ▶ Goal: another researcher can **evaluate and repeat** your work.
- ▶ Previously published method? Give the **reference**. Unpublished? Give **all essential details**.
- ▶ **Keep one voice** within a paragraph: do not switch between passive (“was programmed”) and “we”.
- ▶ Methods is the one section where passive voice still dominates; do not start every sentence with “We”.
- ▶ Ask many colleagues to read it: missing links are invisible to you.

Rule 4: Results, clear, concise, objective

- ▶ Organise by sub-sections: purpose, approach, data (text + visuals), short commentary.
- ▶ **Do not repeat** every number from the table; highlight the most important points.
- ▶ Distinguish **results from data**. You may have 20 runs, but show only what the argument needs. Selecting is fine; distorting is falsification.
- ▶ If a finding contradicts your idea, say so and offer a plausible explanation.
- ▶ Many journals combine Results and Discussion; otherwise keep interpretation for the Discussion.

Cut empty words

Intensifiers (“clearly,” “obviously,” “very,” “basically”) add words and lower credibility.

✗ *Table 3 clearly shows that. . .*

✓ *Table 3 shows that. . .*

Nominalisations bury verbs.

✗ *In this paper we provide an argument that. . .*

✓ *In this paper we argue that. . .*

Cut clearly and obviously.



Rule 5: Introduction, three moves

Move 1: Establish territory

Show the area is important or problematic; review prior work.

Move 2: Find a niche

Indicate a **gap** in previous work, or extend it.

Move 3: Occupy the niche

State purpose, hypotheses, main findings, value, paper structure.

- ▶ Zoom in from **general to specific**. Typically only about **250–600 words**, yet it is often the hardest section.
- ▶ No “suspense”: give a one-sentence summary of results and approach. The reader should not hunt for them later.
- ▶ Never assume the importance is obvious; **state the benefit** of your work explicitly.
- ▶ Re-read Methods and Results first and update the outline; your focus may have drifted.

Tell the reader what is new, early.



Rule 6: Discussion, zoom back out

Move 1: Major findings

State them; explain meaning and importance; consider alternative explanations.



Move 2: Research context

Compare with other work; explain discrepancies; state **limitations**.



Move 3: Close

Answer the research questions; state applications and implications.

- ▶ Open with the major finding that answers the question from your Introduction: “Our findings demonstrate...”
- ▶ Never assume the significance is obvious; **say why it matters**.
- ▶ Address likely objections yourself. It earns trust and avoids wrong interpretations.
- ▶ End with a **take-home message**. Use clear sentences, active voice, and a confident but modest tone.

End with a take-home message.



- ▶ Organise by **themes or ideas**, not as a list “Smith did X. Jones did Y.”
- ▶ End each theme with the **gap** that motivates your work.
- ▶ Cite **primary sources**; do not cite a paper you have not opened.
- ▶ Use reference software from day one (Zotero, EndNote, BibTeX/JabRef). Retro-fitting citations is painful.

Quick rules

- ✓ Quote or paraphrase **with** citation
- ✓ Give credit for datasets, code, and figures
- ✓ Obtain permission for reused images
- ✗ “Citation padding” with irrelevant papers
- ✗ Copying sentences from papers, even with a small word swap
- ✗ Citing a source you only know from an AI summary

Cite only what you have opened.



An abstract should...

- ▶ **Summarise** the article; it is not an introduction
- ▶ **Tell a story**: a puzzle or problem, not a barrage of data
- ▶ State the **argument** and why it matters
- ▶ Reveal the most valuable **findings** up front
- ▶ State methods briefly, **one sentence**
- ▶ Use **strong verbs**: “argues,” “demonstrates,” “shows,” not “explores,” “attempts to”
- ▶ Include the most relevant **keywords**
- ▶ Be **self-contained**
- ▶ Report what you **did** (past), not what you hope to do

Usually avoid in an abstract

Footnotes, citations, quotations, abbreviations, symbols and acronyms (spell out all terms).

Why it matters

Many readers never look beyond the abstract; make it so good that someone could cite your article accurately from it alone. One editor: seeing “explores” in an abstract is a red flag that there is no argument.

Fill-in-the-brackets example (illustrative)

Background: [Problem in one sentence; why it matters.]

Objective: [Gap and what this study set out to do.]

Methods: [Proposed method, data, and evaluation, in one sentence.]

Results: [Key quantitative result vs. the strongest baseline: $X\%$ vs. $Y\%$ on dataset D .]

Conclusion: [What the result means; scope and limitation.]

Keywords: [5–6 searchable terms]

- ▶ Structured journals give you **5–8 sub-headings**; the five common ones are background, objective, method, results, discussion/conclusion.
- ▶ Even when unstructured, use the **same five moves** as sentences.
- ▶ Stay within the **word limit**; portals often reject longer abstracts.
- ▶ Ask a friend outside your project: *can you tell what I found?*

- ▶ **Name your subjects:** model, task, data, population, setting.
- ▶ Avoid **broad** titles (under about seven words is a warning sign) and strings of vague abstractions.
- ▶ Put **searchable keywords** in the title; around 60% of articles are found through Google-type searches.
- ▶ Use the commonly used term (check Google Scholar hit counts), singular and plural.
- ▶ Make it **coherent**: put keywords in relation using prepositions; avoid comma-strings.
- ▶ Longer titles tend to be cited more; titles with a colon often do better; question titles do worse.

Weak

Deep Learning for Medical Images

Better (illustrative)

Boundary-Aware U-Net Loss for Lesion-Edge Segmentation in Low-Contrast Dermoscopy Images

The better title names the method, task, data type, and implied contribution, and it contains the words someone would actually search.

- ▶ Each figure/table must be **self-explanatory**: clear caption, units, legend, readable fonts, vector graphics where possible.
- ▶ Same metric, same order, same decimals across tables; bold the best result honestly.
- ▶ Show **error bars** or SD over runs.
- ▶ Check image **permissions** and dataset licences.
- ▶ Put results in **visuals** and let the text interpret them.

Follow the venue's style

Journals follow a style manual: **IEEE/Vancouver/AMA** (numbered), **APA**, **Chicago**, **MLA**. Three things matter most at submission: **documentation, punctuation, spelling**.

Use a reference manager that can switch styles. Typical errors editors complain about: wrong length, abstract too long, wrong font/spacing, missing keywords, wrong citation style.

PART 4

Revising

Fix the big things first.



Macro-revision (ideas)

Organisation, content, flow, argument. Move, add, delete or rewrite sections.

Work in small chunks (about five pages at a time); work linearly with an outline.

Micro-revision (words)

Sentences, grammar, punctuation, spelling, word choice. Non-linear; you can start from the end or read backwards.

- ▶ Novices spend too much time polishing words. Experienced writers make about three times as many changes that alter meaning than novices.
- ▶ Do **macro first**; polishing sentences you will delete is wasted work.
- ▶ Expect **5–7 drafts**. If you cannot stop revising, set a **deadline**.
- ▶ Good prose cannot rescue bad research, but bad prose can sink good research.

Big fixes first, polish later.



- ▶ **Reverse outline**: write one sentence per paragraph afterwards. Does the story flow? What is missing or repeated?
- ▶ Does every section serve the **argument**? Cut what does not.
- ▶ Does the Discussion answer the question posed in the Introduction?
- ▶ Are the **abstract, introduction and conclusion** telling the same story?

The “So what?” test

Show each key paragraph to a friend and ask: *So what? Why should a reader care?* If you cannot answer in one sentence, the paper is missing a **claim for significance**.

Get non-specialist feedback

Listen patiently. You are not expected to defend your writing; decide later what to change.

Principle	Rule	Example fix
Brevity	Don't use two words where one will do	<i>"blocked and obstructed" → one word</i>
Vigor	Don't use a noun when you can use a verb	<i>"give an explanation of" → "explain"</i>
Potency	Don't use a weak verb when a strong one exists	<i>"is a student of" → "studies"</i>
Dynamism	Avoid passive unless the subject is unknown or unimportant	<i>"was designed by Decq" → "Decq designed"</i>
Lucidity	Don't use a pronoun when a noun is clearer	<i>"it. . . it" → name the noun</i>
Efficiency	Don't use a preposition when a noun or verb will do	<i>chains of "of the. . . of the. . ."</i>
Leanness	Don't use an adjective or adverb unless you must	<i>drop "very," "clearly"</i>
Specificity	Don't use a general word when a specific one exists	<i>"results" → "F1 rose from 0.81 to 0.86"</i>

- ▶ **Read aloud**, to yourself, a friend, or a recorder; you will hear where fluency breaks.
- ▶ **Know your idiosyncrasies**: list your habitual problems and **search** for them (“this/these/those” without a noun, “very,” “really,” “there is/are”).
- ▶ **Print** it: double-spaced, 14-pt, and read line by line with a sheet of paper covering the rest.
- ▶ Share with friends, colleagues, mentor, and writing centre; **non-specialists** are the most honest readers.
- ▶ Revise sections in short **five-page** chunks to keep your eyes fresh.
- ▶ Finish: you will never feel it is perfect. Peer review is part of the improvement process.

PART 5

Submit & review

Rejection is part of the process.



Follow the journal, not your habits

- ▶ Read the **author guidelines** fully; use the official template
- ▶ Page and word limits; abstract length; keywords; article type
- ▶ Reference style; punctuation; US/UK spelling
- ▶ Figures at required resolution and format
- ▶ Author names, affiliations, ORCID, corresponding author

Declarations (often mandatory)

- ▶ Ethics approval and consent; dataset licences
- ▶ Data and code availability
- ▶ Conflicts of interest and funding
- ▶ Similarity (plagiarism) report is clean
- ▶ **Generative-AI use disclosed** if required
- ▶ Do not suggest reviewers unless the journal asks

Don't polish forever. *"If any of the journal's guidelines contradict my advice... follow the journal, not me!"*

- ▶ Address the editor by name; one page at most.
- ▶ Title, article type, **one-sentence argument** and why it matters.
- ▶ Why it fits **this** journal's scope and readership (point to related papers it published).
- ▶ Statement that the work is original and **not under review elsewhere**.
- ▶ Declarations: AI use, conflicts of interest, permissions.

Even in a web portal

Portals often have no cover-letter field, but a short note to the editor still helps.

One page. One-sentence argument.



- ▶ You should get an acknowledgement at once. A week of silence after email submission: send a short query.
- ▶ First decision commonly takes **a few weeks to several months**; reviews average about 3–4 months. Do **not** email after a week.
- ▶ Status check after the stated time. Keep the same polite wording; if there is no reply after repeated enquiries, consider withdrawing.
- ▶ Rejections can arrive within days, either for fit or backlog.

Possible decisions

Accept as is: almost never (about 1% of first submissions in one large study).

Minor revision: celebrate, revise quickly, re-submit.

Major revision / revise & resubmit: you are still in the game. Most papers are published this way.

Reject: revise using the feedback and send elsewhere.

Be patient. Reviews take months.



- ▶ Don't open the email right before class or a meeting. **Wait until you have time** and emotional space; set the reports aside for a day or two.
- ▶ Reviews are **subjective** and sometimes rude, unclear, or even wrong. But authors consistently report that **revising makes the paper stronger**.
- ▶ Reviewers may skim: average reading time is a few hours. Write for the slightly distracted reader.
- ▶ Conflicting reviews: you choose which advice to follow, but **explain** your choice.
- ▶ Assume suggestions are **directives**; you still have choices.

The “golden rule” (Starbuck)

“No reviewer is ever wrong.” Patently untrue, but useful: each reviewer shows how a real reader read your paper. Leap from their comments; don't bow to them.

Persistence wins

How you respond to decisions shapes your career: those who persevere, not necessarily the most brilliant, get published.

Structure

1. Thank the editor and reviewers (briefly, sincerely)
2. **Number every comment**; quote it
3. Reply directly under each: what you changed and **where** (page/line), or why you did not, with **evidence**
4. Mark changes in the manuscript (colour or tracked changes)
5. Keep a polite, factual tone, even if the comment is blunt

If rejected

Wait a few days, read the reviews again, revise, and send to the next venue. Don't resubmit the same unrevised file: the next reviewer may be the same person.

Don't

Argue emotionally, ignore comments silently, or promise experiments you will not run.

Answer every comment politely.



A 12-week plan for your first paper (after Belcher)

1. **Plan:** schedule writing, choose the paper, set up citation software, settle authorship
2. **Argument:** state it in 1–2 sentences, test it, reorganise around it
3. **Abstract:** draft with the five moves; get feedback
4. **Journal:** shortlist and evaluate venues
5. **Literature:** tidy citations; write or revise the related-work review
6. **Significance:** “So what?” and claims of significance
7. **Analyse evidence:** quality, relevance, interpretation
8. **Present evidence:** tables, figures, permissions
9. **Structure:** reverse-outline and reorder
10. **Open and close:** title, introduction, conclusion
11. **Edit sentences:** run the editing diagnostic
12. **Send!** Journal style, cover letter, submit, celebrate

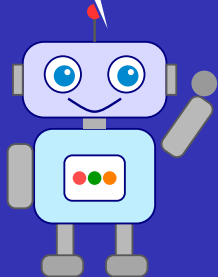
Adapt it

Compress or stretch the weeks to fit your deadline. The point is **steady progress and finishing**: decide what is “beyond the scope of this article,” and **send it**.

PART 6

AI tools

Use me wisely, and always
check my work!



- ▶ A language model generates **plausible text**; it does not check truth, and it can invent facts, numbers and citations with total confidence.
- ▶ **Authorship means accountability**. Every major publisher and body (e.g. ICMJE, COPE, ACM, Elsevier, Springer Nature) says AI cannot be an author because it cannot be accountable.
- ▶ You must be able to **defend every sentence**, every number and every reference in a viva, a rebuttal or a retraction notice.

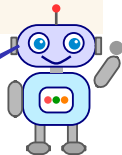
A simple test

If an examiner or reviewer asked, “Show me where this claim comes from,” could you do it **without** asking the chatbot?

Think of AI as...

A fast, tireless, **unreliable** intern: useful for drafts and ideas, never trusted for facts without checking.

I am a tool, not your co-author!



Stage	Reasonable use	Your responsibility
Ideas	Ask for questions to sharpen your outline; brainstorm limitations or objections	Decide the argument yourself
Reading	Get a plain-language explanation of a concept; list search terms	Read the actual paper before citing it
Language	Fix grammar and clarity in your own paragraphs (valuable if English is not your first language)	Check no meaning changed
Structure	Check a draft against a checklist (e.g. abstract has background, method, result)	Fix issues in your own words
Code and LaTeX	Boilerplate, plotting scripts, BibTeX formatting, debugging hints	Test and understand every line
Rehearsal	Mock reviewer questions on your <i>own</i> draft (if it is not confidential)	Answer from your evidence

Never do this

1. **Cite references the model gave you** without opening them. They may not exist.
2. **Let it write your argument, results or conclusions.** It cannot know what your experiments showed.
3. **Generate or “improve” data, figures or images** in a research paper.
4. **Paste confidential material:** unpublished results, patient data, others’ manuscripts.
5. **Submit AI text as your own** where disclosure is required.

Quieter damage

- ▶ **Hallucinated facts and numbers,** presented in fluent prose
- ▶ **Paraphrase without attribution:** still plagiarism
- ▶ **Generic voice:** every paper sounds the same; reviewers notice
- ▶ **Subtle meaning drift:** “may” becomes “does,” a limitation disappears
- ▶ **Outdated or biased knowledge:** misses recent baselines
- ▶ **Skill atrophy:** you never learn to write or argue

- ▶ Chatbots produce references that look perfect (plausible authors, journal, volume, DOI) **but do not exist** or do not say what is claimed.
- ▶ A well-known cautionary case: in *Mata v. Avianca* (US federal court, 2023), lawyers were sanctioned after filing a brief with cases a chatbot had made up.
- ▶ Publishers explicitly hold authors responsible for **fabricated references**.
- ▶ Fake citations are easy to detect: reviewers and editors look them up.

Verify every reference

1. Does the **DOI/URL** open the *same* paper? (Crossref, publisher site)
2. Do **authors, title, venue, year** match exactly? (Google Scholar, Scopus, PubMed)
3. **Open the paper** and find the passage that supports your claim.
4. Add it to your reference manager from the **publisher/database** record, not from chat text.

Rule

If you haven't opened it, don't cite it.

Topic	Common position (check your venue's current policy!)
Authorship	AI cannot be listed as an author or co-author (ICMJE, COPE, ACM, Elsevier, Springer Nature, Oxford, Cambridge, and others)
Disclosure	Required by most (Methods, Acknowledgments, or a declaration). ACM: proportionate to AI-generated content; Taylor & Francis: all AI use; Elsevier, Springer Nature, Sage: plain grammar fixes exempt; IEEE: in Acknowledgments
Images & data	Elsevier bars generative AI for creating or altering research images; Springer Nature generally does not permit AI images; Taylor & Francis bars it for core research data
Peer review	Reviewers must not upload manuscripts into public AI tools (confidentiality)
Responsibility	Named humans answer for accuracy, references and images

Do not paste into public chatbots

- ▶ Unpublished results, code or manuscripts (yours or others')
- ▶ **Patient data**: images, reports, identifiers; even "anonymised" clinical data may be restricted by hospital ethics approvals and data-protection law
- ▶ Datasets under a data-use agreement or NDA
- ▶ Anything from **peer review**: reviewing is confidential
- ▶ Funder, company, or supervisor-owned material without permission

- ▶ Many tools may **retain inputs or use them to improve models**, depending on plan and settings. Read the data-controls page.
- ▶ Prefer **institution-approved** tools or local models for sensitive text.
- ▶ When in doubt, **ask your supervisor** before sharing.
- ▶ Strip names, IDs, affiliations from any text you do share.
- ▶ In double-blind submissions, pasting your manuscript into a tool with search or sharing features can also leak authorship.

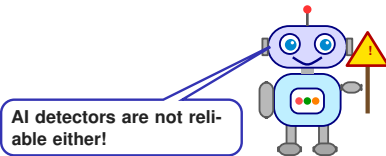
Elsevier and others cite confidentiality and proprietary-rights concerns about uploading manuscripts. Data-protection practice: consult your institution's ethics board and policy.

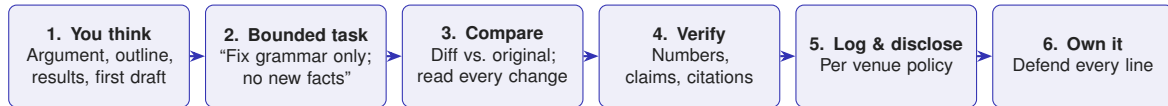
- ▶ Journals and institutions run **similarity checks** (e.g. iThenticate, Turnitin). Lightly reworded copying is still caught or judged as plagiarism.
- ▶ **AI detectors are unreliable**. They produce false positives, and a 2023 study (Liang et al., *Patterns*) found that they **misclassify writing by non-native English speakers** far more often.
- ▶ So don't “fix” a draft with a humaniser tool: it harms quality and is itself deceptive.

Liang et al., “GPT detectors are biased against non-native English writers,” *Patterns* 4(7), 2023.

Protect yourself with a paper trail

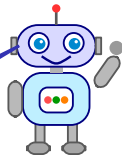
- ▶ Keep **dated drafts** and version history
- ▶ Keep notes, raw data, scripts, experiment logs
- ▶ Log AI use: tool, version/date, purpose, what you changed
- ▶ Be ready to explain your paper orally and draw the key figure from memory





- ▶ **Draft first, then ask AI** (not the reverse). Your thinking stays yours.
- ▶ Give **narrow instructions**; paste only non-confidential text.
- ▶ Always compare output with your original: look for changed numbers, hedges, and dropped limitations.
- ▶ Verify **independently** (database, original paper, your logs), not by asking the chatbot again.
- ▶ Ask your supervisor and co-authors; some groups have written AI policies.
- ▶ If unsure whether to disclose, **disclose**.

You think first. I help second.



✗ Risky prompt

“Write the related work for my paper on X with citations.”

“Give me 20 references on X.”

“Write my abstract.”

“Make my results sound impressive.”

“Improve my paper.”

✓ Safer prompt

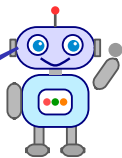
“Here are the 8 papers I read and my notes. Suggest 3 themes to group them. Do not add any references.”

“Suggest search keywords and sub-topics for X that I can search in Scopus and Google Scholar.”

“Here is my abstract. Check whether it contains background, objective, method, result and conclusion; list what is missing. Do not rewrite.”

“Shorten this results paragraph by 20%. Do not change any numbers, hedges, or claims; list every change.”

“Fix grammar and clarity in this paragraph. Keep technical terms, numbers and citations exactly as they are. Show changes as a table.”



Be specific. Narrow prompts win.

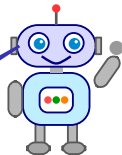
Template (adapt to the journal's wording)

"During the preparation of this work the author(s) used [TOOL, VERSION] to [PURPOSE, e.g. improve the language and readability of the text]. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication."

Template wording follows the statement recommended by Elsevier for the writing process; other publishers use different wording, so check each venue's instructions.

- ▶ Place it where the venue asks: Methods, Acknowledgments, or a declaration section; mention it in the cover letter if asked.
- ▶ Name the **tool, version, date and purpose**.
- ▶ AI is **not** listed as an author, and is not cited as a source of facts.
- ▶ Disclosure protects you; hiding use is what causes retractions.

Tell them you used me.

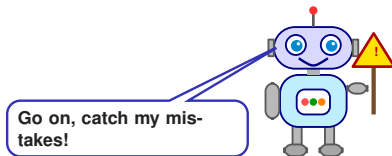


Class activity: catch the hallucination (10 minutes)

1. In pairs, ask a chatbot: “List five key papers on [your project topic] with DOIs.”
2. For **each** reference, check: does the DOI resolve? Do authors, title, venue, year match? Does the paper really contain the claim?
3. Record how many were **real**, **partly wrong**, or **fabricated**.
4. Share one example with the class.

Discuss

- ▶ Which errors were hardest to spot?
- ▶ What would have happened if you had cited them?
- ▶ How would you change your workflow?



✓ Do

- ▶ Draft your own argument, results and structure first
- ▶ Use AI for language polish, brainstorming and checklists
- ▶ Verify every fact, number and reference independently
- ▶ Keep a log; keep your drafts
- ▶ Disclose per the venue's policy
- ▶ Ask your supervisor when unsure

✗ Don't

- ▶ Cite what you haven't opened
- ▶ Let AI invent data, figures, or conclusions
- ▶ Paste confidential, patient, or peer-review material
- ▶ List AI as an author or hide its use
- ▶ Use detectors or "humanisers" to disguise AI text
- ▶ Submit text you can't explain

PART 7

Wrap-up

Almost there. Write it, revise it, send it!



Content

- ☐ Argument stated in 1–2 sentences, early
- ☐ Every section serves the argument
- ☐ Baselines, ablations, variance reported
- ☐ Limitations stated honestly
- ☐ Methods allow replication; code/data statement
- ☐ Title, abstract, intro, conclusion tell one story

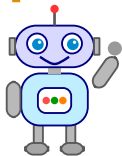
Integrity and format

- ☐ Every reference opened, verified, and in the right style
- ☐ Similarity report checked; no copied text
- ☐ Permissions, ethics, licences, conflicts declared
- ☐ Authorship and order agreed by all
- ☐ AI use logged and disclosed as required
- ☐ Venue template, limits, and cover letter done

1. A paper is an **argument**: one new idea, clearly stated and clearly related to existing work.
2. Choose the **venue** deliberately and avoid predatory outlets; never submit to two venues at once.
3. **Schedule, outline, draft, then revise** (macro first, micro later); write for the skimmer.
4. Expect revision and rejection. **Persistence** gets papers published.
5. AI is a **tool, not a co-author**: draft first, verify everything, protect confidential data, and disclose.

One sentence to remember

Use AI to polish your thinking, never to replace it.



You have got this. Go write!



- ▶ Belcher, W. L. (2019). *Writing Your Journal Article in Twelve Weeks: A Guide to Academic Publishing Success*, 2nd ed. University of Chicago Press.
- ▶ Kallestinova, E. D. (2011). How to write your first research paper. *Yale Journal of Biology and Medicine*, 84(3), 181–190.
- ▶ Silvia, P. J. (2007). *How to Write a Lot*. American Psychological Association.
- ▶ Swales, J. M., & Feak, C. B. *Academic Writing for Graduate Students*. University of Michigan Press.
- ▶ Liang, W., et al. (2023). GPT detectors are biased against non-native English writers. *Patterns*, 4(7).

Publisher policies change frequently; always check the current author instructions of your target venue.

Questions?

Write it. Revise it. Send it.

